

Data Warehouses and Data Mining Systems

Report no. 3

Exercise topic:

Phase 3: Logical modelling

Project topic:

Counter Strike skin market analysis

Section members:

Mateusz Znalezniak

Tymoteusz Kłoska

Gliwice, 30.03.2026

1 Data Warehouse Datasheet

1.1 Fact Table

Fact: Market Price

- **Measures:**
 - price (float)
 - volume (int)
- **Granularity:** one record per item per timestamp
- **Time Variability:** fully time-variant (historical data is preserved)
- **Description:** stores transactional market data representing item prices and traded volumes over time
- **Source Systems:**
 - Market transaction datasets
 - Game item metadata (skins, stickers, weapons)
 - Container datasets
 - Esports match results datasets

1.2 Dimensions

1.2.1 Time

- Attributes: date, day, month, year
- Description: provides temporal context for each transaction

1.2.2 Skin

- Attributes: skin_id, skin_name, rarity
- Description: represents weapon skins available on the market

1.2.3 Sticker

- Attributes: sticker_id, sticker_name, rarity
- Description: represents collectible stickers associated with teams or events

1.2.4 Weapon

- Attributes: weapon_id, weapon_name, weapon_type
- Description: describes the base weapon associated with skins

1.2.5 Container

- Attributes: container_id, container_name, container_price, release_date, container_type
- Description: represents cases or containers from which items are obtained

1.2.6 Team

- Attributes: team_id, team_name
- Description: represents esports teams related to stickers or matches

1.2.7 Match Outcome

- Attributes: match_id, match_date, winner_team, loser_team
- Description: stores results of professional matches

1.2.8 Price Range

- Attributes: price_range_id, price_range
- Description: categorizes items into price intervals for analysis

1.2.9 Wear Range

- Attributes: wear_range_id, wear_float, wear_range
- Description: categorizes item condition based on wear values

1.3 Relationships

- The schema follows a **star schema** design
- The fact table is connected to all dimensions via foreign keys
- All relationships are **many-to-one (N:1)** from fact to dimensions
- No hierarchical normalization (no snowflaking)

2 Logical Model

The logical model follows a **star schema**, where all dimensions are directly connected to the fact table without normalization.

2.1 Entities

Fact

- Market Price (price, volume)

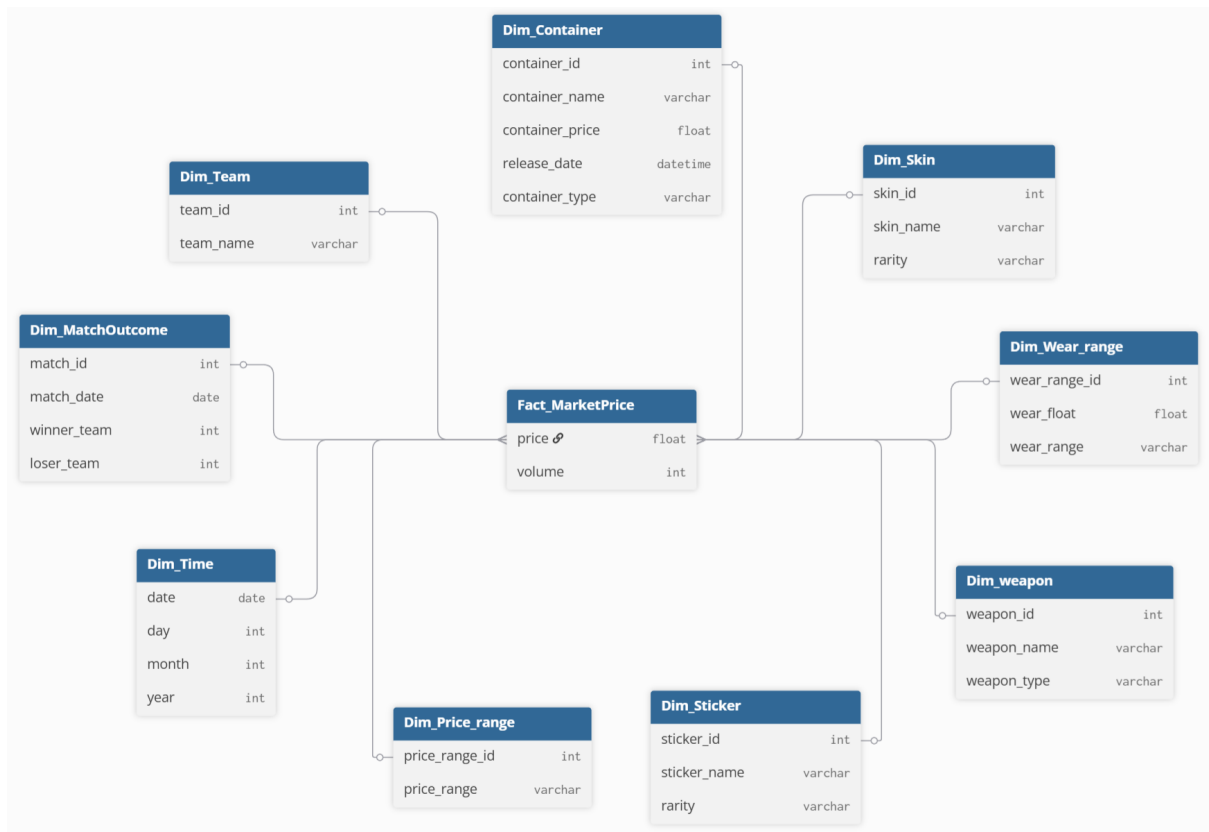
Dimensions

- Time
- Skin
- Sticker
- Weapon
- Container
- Team
- Match Outcome
- Price Range
- Wear Range

2.2 Relationships

- Fact $\rightarrow$ all dimensions (N:1)
- No hierarchical relationships (fully denormalized dimensions)

2.3 Star Schema Diagram



3 Conclusion

The final model adopts a star schema structure, ensuring efficient analytical processing of market price data. By denormalizing all dimensions, the design prioritizes query performance and simplicity, making it well-suited for large-scale analysis of game item markets and esports-related data.